31.3: The ancestor of fungi was an aquatic, single-celled, flagellated protist

Data from molecular systematics offer insights into the early evolution of fungi. As a result, systematists now recognize that fungi and animals are more closely related to each other than either group is to plants or to most other eukaryotes.

The Origin of Fungi

Phylogenetic analyses suggest that fungi evolved from a flagellated ancestor. While the majority of fungi lack flagella, two basal lineages of fungi (the cryptomycetes and the chytrids, as we'll discuss shortly) do have flagella. Moreover, most of the protists that share a close common ancestor with animals and fungi also have flagella. DNA sequence data indicate that these three groups of eukaryotes—the fungi, the animals, and their protistan relatives—form a monophyletic group, or clade **(Figure 31.8)**. As discussed in Concept 28.5, members of this clade are called ****opisthokonts**, a name that refers to the posterior (*opistho-*) location of the flagellum in these organisms.

Figure 31.8

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Fungi and their close relatives.
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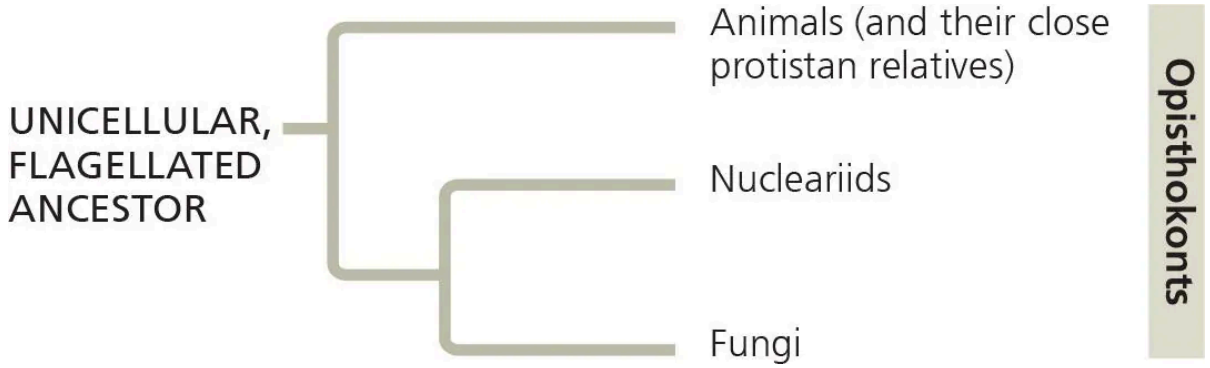


Molecular evidence indicates that the nucleariids, a group of single-celled protists, are the closest living relatives of fungi.

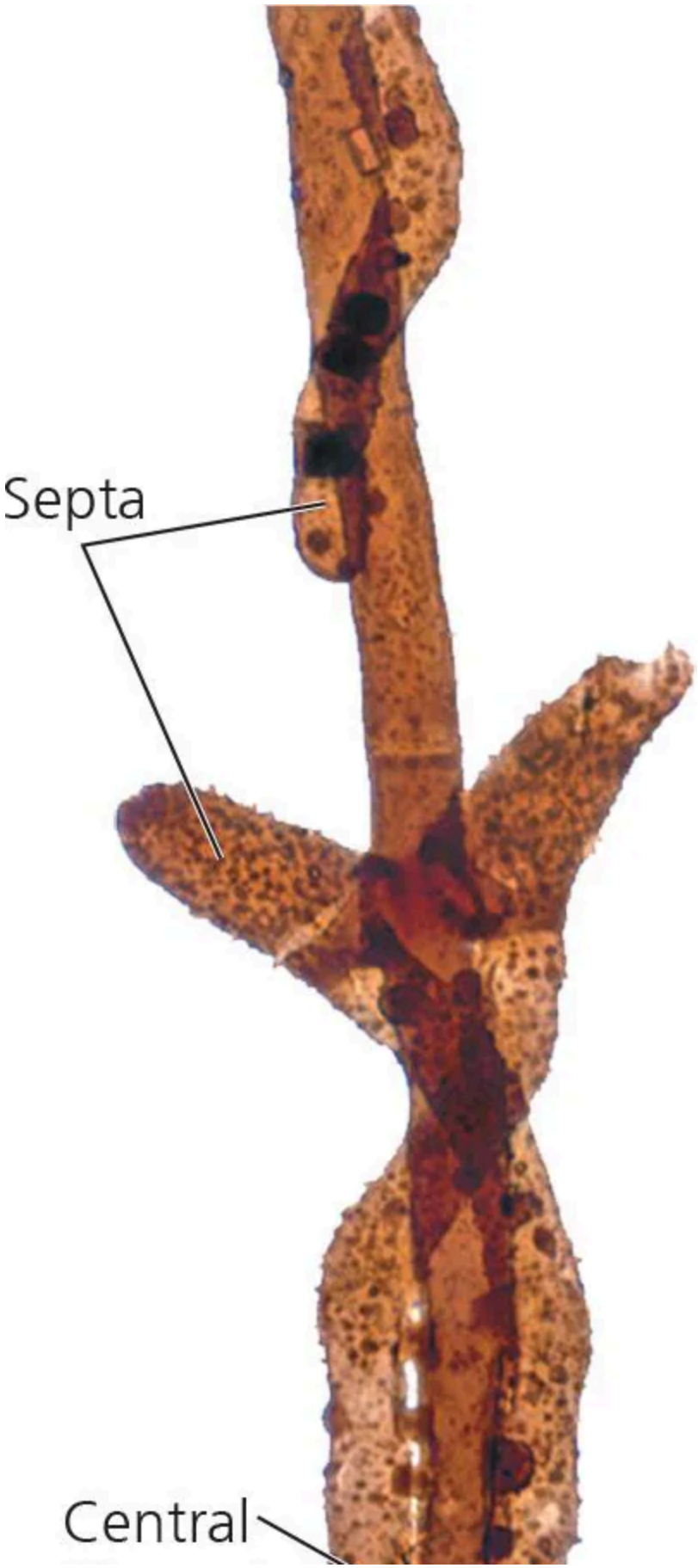
The tree begins with a unicellular, flagellated ancestor. One branch goes to animals (and their close protistan relatives). Another branch goes to a second branch. That branch goes to 1) nucleariids and 2) fungi.

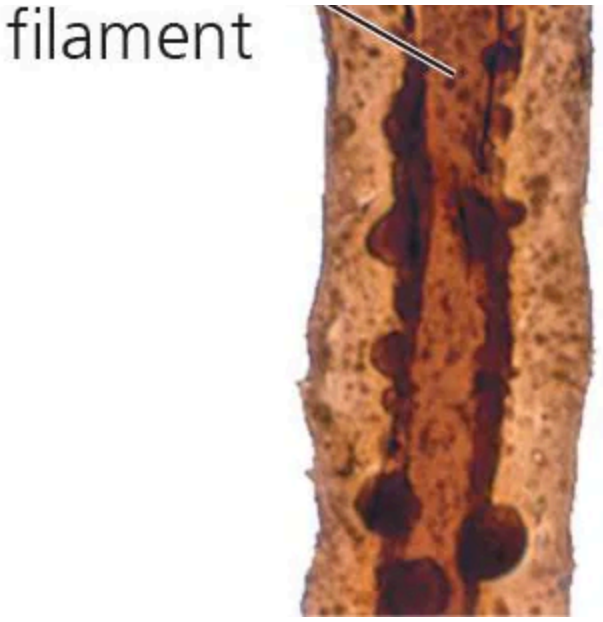
Within the opisthokont clade, fungi are more closely related to several groups of single-celled protists than they are to animals, suggesting that the ancestor of fungi was unicellular. One such group of unicellular protists, the **nucleariids**, consists of amoebas that feed on algae and bacteria. DNA evidence further indicates that animals are more closely related to a *different* group of protists (the choanoflagellates) than they are to either fungi or nucleariids. Together, these results suggest that multicellularity evolved in animals and fungi independently, from different single-celled ancestors.

Using molecular clock analyses, scientists have estimated that the ancestors of animals and fungi diverged into separate lineages more than a billion years ago. Fossils of certain multicellular, marine eukaryotes that lived 1 billionyears ago have been interpreted as fungi, but some scientists have questioned this conclusion. Furthermore, although fungi probably originated in aquatic environments, the oldest fossils that are widely accepted as fungi are of terrestrial species that lived 440 millionyears ago **(Figure 31.9)**. Fungi may have colonized land as early as 635 millionyears ago: Terrestrial fossils of that age were interpreted as “probable fungi” in a 2021 study. Overall, more fossils are needed to help clarify when fungi originated, what features were present in their earliest lineages, and when they first colonized land.

Figure 31.9

Fossil hyphae from the fungus *Tortotubus* (440 millionyears ago).


Septa
Central



The central filament is surrounded by two partially overlapping filaments (LM).

The Move to Land

Plants colonized land about 470 millionyears ago (see Concept 29.1), and fungi may well have colonized land before plants. Indeed, some researchers have described life on land before the arrival of plants as a “green slime” that consisted of cyanobacteria, algae, and a variety of small, heterotrophic species, including fungi. With their capacity for extracellular digestion, fungi would have been well suited for feeding on other early terrestrial organisms (or their remains).

Once on land, some fungi formed symbiotic associations with early plants. For example, 405-million-year-old fossils of the early plant *Aglaophyton* contain evidence of mycorrhizal relationships between plants and fungi (see Figure 25.13). This evidence includes fossils of hyphae that have penetrated within plant cells and formed structures that resemble the arbuscules formed today by arbuscular mycorrhizae. Similar structures have been found in a variety of other early plants, suggesting that plants probably existed in beneficial relationships with fungi from the earliest periods of colonization of land. The earliest plants lacked roots, limiting their ability to extract nutrients from the soil. As occurs in mycorrhizal associations today, it is likely that soil nutrients were transferred to early plants via the extensive mycelia formed by their symbiotic fungal partners.

Support for the antiquity of mycorrhizal associations has also come from molecular studies. For a mycorrhizal fungus and its plant partner to establish a symbiotic relationship, certain genes must be expressed by the fungus and other genes must be expressed by the plant. Researchers focused on three plant genes (called *sym* genes) whose expression is required for the formation of mycorrhizae in flowering plants. They found that these genes were present in all major plant lineages, including basal lineages such as

liverworts (see Figure 29.13). Furthermore, after they transferred a liverwort *sym* gene to a flowering plant mutant that could not form mycorrhizae, the mutant recovered its ability to form mycorrhizae. These results suggest that mycorrhizal *sym* genes were present in early plants—and that the function of these genes has been conserved for hundreds of millions of years as plants continued to adapt to life on land.